

Siwani Sah

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SUMMARY

Software Engineer with 3 years of production experience building scalable enterprise applications and AI-driven systems. Specializing in full stack architecture, microservices, and applied AI.

EXPERIENCE

Florida Polytechnic University

Graduate and Research Assistant

Lakeland, Florida

Aug 2025- Current

- Developing an AI code review system that pairs multiple AI models based on behavioral differences and applies execution-based verification, which catches bugs that a single model misses and improves overall defect detection accuracy.
- Led project supervision and graded cohorts of 50+ students for Computer Architecture/ Computation & Programming (CDA3100/CIS4339), ML (CAP 4612).

Projects: Intelligent Sprint Analysis (Repo-Sprint), Autonomous Self-Healing LLM Unit Test Generator, Diverse Strings - Max Hamming Distance in a Σ -DAG.

Professional Computer System (P) Ltd

Software Engineer

Kathmandu, Nepal

Jul 2022-Jul 2025

- Led full-stack development of enterprise web applications for Nepal Tourism Board (NTB) and Institute of Chartered Accountants of Nepal (ICAN) ERP, supporting 20,000+ members with .NET Core, React.js, PostgreSQL, and a microservices architecture.
- Designed and implemented APIs, database schemas, and React components that improved data processing speeds by 25% and reduced manual data entry by 30%.
- Optimized complex SQL queries, indexing strategies, and caching mechanisms across microservices, reducing database response times by 35%.
- Built document intake and validation workflows that reduced manual entry errors by 30%.
- Improved frontend performance and UX, cutting page load times by 40% and reducing user-reported issues by 15%.
- Deployed and maintained services on AWS EC2 and IIS, strengthening cloud network security and ensuring 91.9% application availability and a 20% increase in system performance.

Software Engineer Trainee / Intern

Dec 2021-Jul 2022

- Contributed to the successful delivery of the Nepal Election Commission ERP project, resolving over 30 critical bugs before launch by assisting in design, debugging, and client support using C#, .NET, Knockout.js, and Oracle DB.
- Built a full-stack system to track employee records and qualifications using .NET 8 and PostgreSQL, with a Knockout.js frontend that makes data entry fast and interactive.
- Enhanced user experience by debugging and improving UI, resulting in a 15% reduction in user-reported issues.

TECHNICAL SKILLS

Programming Languages: C#, C, Python, JavaScript/TypeScript, HTML, CSS.

Frameworks & Libraries: .NET (Core, MVC, Web Forms), React.js, Node.js, Knockout.js.

AI & Machine Learning: LLMs, Multi-Agent Systems, AI Code Verification.

Methodologies & Concepts: Agile/Scrum, SDLC, OOP, Microservices, Full Stack Development.

Databases: PostgreSQL, MySQL, Oracle.

Tools & Platforms: Git/GitHub, Visual Studio, LaTeX.

Infrastructure & DevOps: Docker, Internet Information Services (IIS), Linux/Unix, AWS, REST APIs.

EDUCATION

Florida Polytechnic University, Florida, USA

Aug 2025-May 2027

Master's, Computer Science

Relevant Courses: Computer Network Programming, Software Engineering, Advanced Algorithm design, Network Security, Machine Learning, AI.

PROJECTS

Agent Flow AI Assistant: Llama, JavaScript, Cloudflare, RAG

- Architected a serverless AI assistant by designing an autonomous agent capable of dynamic tool execution for real-time web search.
- Built a scalable RAG engine using AWS S3 and vector databases to retrieve and process context-aware documents and query performance.
- Integrated multimodal vision models for image analysis and implemented Cloudflare Durable Objects to manage persistent, real-time conversation state.

Institute of Chartered Accountants of Nepal (ICAN) ERP: Full Stack Developer | C#, .NET Core, NodeJS, React.js, PostgreSQL

- Improved operational efficiency and data accuracy for over 20,000 members by designing, developing, and deploying enterprise modules within a microservices architecture.
- Coordinated directly with clients, translated requirements into scalable solutions and delivered within deadlines.
- Ensured high reliability and performance by deploying and maintaining applications on Windows Server (IIS), overseeing security and updates.

Nepal Tourism Board (NTB) ERP: Full Stack Developer | C#, .NET Core, Knockout.js, MySQL

- Developed and enhanced tourism-related application modules across frontend and backend layers.
- Improved application stability and user experience by implementing MySQL database optimizations, bug fixes, and new support features.
- Delivered timely updates and new features on schedule by collaborating directly with NTB stakeholders to translate requirements into technical solutions.